

SE LAB1 USE CASE – 2 (Dhriti Kamani, PES1UG24AM085)

Lab 1: Requirements Engineering & UML Use-Case Modelling | Problem Statement #07 — Campus & Academic Operations

UC-02: Assign Ticket & SLA (FR-002)

Primary Actor: Maintenance Warden

Supporting Actors: Maintenance Staff

Preconditions: Ticket exists in unassigned queue with 'Open' status.

Postconditions (Success): Ticket status updated to 'Assigned', staff notified.

Main Success Scenario

- Warden logs into admin portal and navigates to 'Unassigned Ticket Queue'.
- System displays list of newly submitted tickets sorted by severity (High priority on top).
- Warden opens a ticket, reviews description, category, and room location.
- Warden selects an available Maintenance Staff member from department roster.
- System automatically populates recommended SLA target resolution timeframe based on issue severity.
- Warden confirms assignment and submits.
- System updates ticket status to 'Assigned', records warden assignment timestamp, and sends instant push/SMS notification to assigned staff member. Use case ends.

Alternate Flow — 4a: Staff Unavailable

- Triggered at step 4 if no maintenance staff are available for the selected category.
- Warden flags ticket for External Vendor Dispatch.
- System logs extended SLA timeline note and notifies Hostel Administration. Flow resumes at step 6.

REQUIREMENTS TABLE

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
1	FR-001	Functional	High	Pass: Ticket is created, assigned a unique ID, and routed to the warden queue. Fail: Incomplete form (e.g., missing room tag) is accepted.	Core intake function; accurate tagging and severity are required to route and prioritize the ticket.
2	FR-002	Functional	High	Pass: Warden selects a staff member from a list, ticket status changes to "Assigned," and the assigned staff is notified within 1 minute. Fail: Ticket remains unassigned after the warden submits the assignment.	Ensures every logged issue is routed to a responsible person for timely resolution.
3	FR-003	Functional	High	Pass: Each status change is timestamped and visible on the resident's ticket view within 1 minute. Fail: A status change is not reflected to the resident.	Keeps residents informed and creates an auditable timeline needed for SLA tracking.
4	FR-004	Functional	Medium	Pass: A ticket cannot transition to Closed without a stored sign-off (resident ID + timestamp). Fail: A ticket is marked Closed with no sign-off record.	Verifies the issue is actually resolved from the resident's perspective and creates accountability for maintenance staff.
5	FR-005	Functional	Medium	Pass: The resident's dashboard lists every submitted ticket with current status and last-updated timestamp. Fail: A submitted ticket is missing from the resident's list.	Transparency reduces duplicate ticket submissions and manual status-check requests to the hostel office.
6	FR-006	Nonfunctional	Medium	Pass: Benchmarking tests confirm target latency and security standards are met under simulated peak load, and escalation alerts fire within the 24-hour window.	Prevents high-severity issues (e.g., electrical hazards) from being silently ignored past the SLA.
7	NFR-001	Performance	High	Pass: Usability testing with 10 residents shows average submission time < 90s. Fail: average exceeds 90s.	Low reporting friction increases adoption and reduces issues going unreported.
8	NFR-002	Security	Medium		

USECASE - DIAGRAM

UML Use-Case Diagram: Hostel Maintenance & Issue Ticketing System

Lab 1: Requirements Engineering & UML Use-Case Modelling | Problem Statement #07

